The human eye provides an excellent example of these concepts. As described in the last chapter, the first layer of the retina transforms an image represented as a pattern of light into an image represented as a pattern of nerve impulses. The second layer of the retina *processes* this neural image and passes it to the third layer, the fibers forming the optic nerve. Imagine that the image being projected onto the retina is a very small spot of light in the center of a dark background. That is, an *impulse* is fed into the eye. Assuming that the system is linear, the image processing taking place in the retina can be determined by inspecting the image appearing at the optic nerve. In other words, we want to find the *point spread function* of the processing. We will revisit the assumption about linearity of the eye later in this chapter.

Figure 24-1 outlines this experiment. Figure (a) illustrates the impulse striking the retina while (b) shows the image appearing at the optic nerve. The middle layer of the eye passes the bright spike, but produces a circular region of increased *darkness*. The eye accomplishes this by a process known as *lateral inhibition*. If a nerve cell in the middle layer is activated, it decreases the ability of its nearby neighbors to become active. When a complete image is viewed by the eye, each point in the image contributes a scaled and shifted version of this impulse response to the image appearing at the optic nerve. In other words, the visual image is *convolved* with this PSF to produce the neural image transmitted to the brain. The obvious question is: how does convolving a viewed image with this PSF improve the ability of the eye to understand the world?

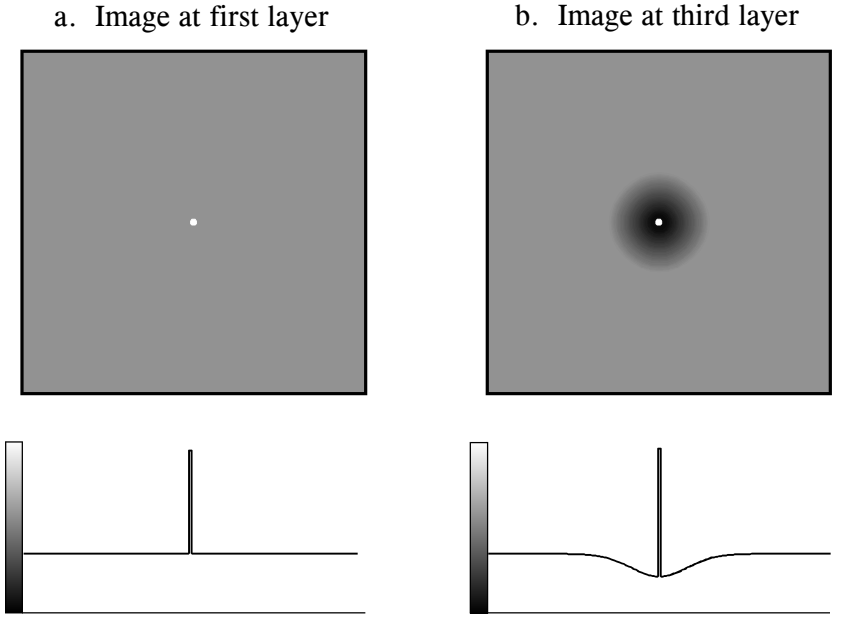

FIGURE 24-1
The PSF of the eye. The middle layer of the retina changes an impulse, shown in (a), into an impulse surrounded by a dark area, shown in (b). This point spread function enhances the edges of objects.